



SQL Interview Q&A

1. What is SQL?

Answer: SQL stands for Structured Query Language, and it is a domain-specific language used for managing and manipulating relational databases.

2. What is a database?

Answer: A database is a structured collection of data that is organized and stored for efficient retrieval and manipulation.

3. Explain the difference between SQL and MySQL.

Answer: SQL is a language used to manage and manipulate relational databases, while MySQL is an open-source relational database management system (RDBMS) that uses SQL as its query language.

4. What is a primary key?

Answer: A primary key is a unique identifier for a record in a database table. It ensures that each record can be uniquely identified and helps maintain data integrity.

5. What is a foreign key?

Answer: A foreign key is a field in a database table that is used to establish a link between the data in two tables. It creates a relationship between the tables by referencing the primary key of another table.

6. Explain the types of SQL commands.

Answer: SQL commands can be broadly categorized into Data Definition Language (DDL), Data Manipulation Language (DML), Data Query Language (DQL), and Data Control Language (DCL).

7. What is the difference between CHAR and VARCHAR data types?

Answer: CHAR is a fixed-length character data type, while VARCHAR is a variable-length character data type. VARCHAR is more flexible as it only stores the characters entered, whereas CHAR pads the remaining spaces with blanks.

8. What is normalization?

Answer: Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It involves dividing large tables into smaller tables and defining relationships between them.

9. Explain the difference between INNER JOIN and LEFT JOIN.

Answer: INNER JOIN returns only the rows where there is a match in both tables, while LEFT JOIN returns all rows from the left table and the matched rows from the right table. If there is no match, NULL values are returned for columns from the right table.

10. What is the purpose of the GROUP BY clause?

Answer: The GROUP BY clause is used to group rows that have the same values in specified columns into summary rows, like finding the total or average for each group.

11. Explain the difference between UNION and UNION ALL.

Answer: UNION combines the result sets of two queries, eliminating duplicate rows, while UNION ALL combines the result sets including duplicates.

12. What is an index, and why is it used?

Answer: An index is a data structure that improves the speed of data retrieval operations on a database table. It is used to quickly locate and access the row in a table based on the values in one or more columns.

13. What is a stored procedure?

Answer: A stored procedure is a precompiled collection of one or more SQL statements that can be executed as a single unit. It is stored in the database and can be called by name.

14. Explain the ACID properties of a transaction.

Answer: ACID stands for Atomicity, Consistency, Isolation, and Durability. It is a set of properties that guarantee that database transactions are processed reliably.

15. What is the purpose of the HAVING clause?

Answer: The HAVING clause is used in combination with the GROUP BY clause to filter the results of a group based on a specified condition.

16. What is a view in SQL?

Answer: A view is a virtual table derived from one or more tables. It does not store the data itself but provides a way to represent the result of a stored query.

17. Explain the difference between a clustered and a non- clustered index.

Answer: A clustered index determines the physical order of data in a table, whereas a non-clustered index does not affect the physical order and creates a separate structure for faster data retrieval.

18. What is the purpose of the SQL ORDER BY clause?

Answer: The ORDER BY clause is used to sort the result set of a query in ascending or descending order based on one or more columns.

19. What is a trigger?

Answer: A trigger is a set of instructions that are automatically executed (or “triggered”) in response to certain events on a particular table or view in a database.

20. Explain the difference between a candidate key, primary key, and a super key.

Answer: A super key is any set of columns that uniquely identifies a row, a candidate key is a minimal super key, and the primary key is the chosen candidate key for a table.

21. What is the purpose of the SQL LIKE statement?

Answer: The LIKE statement is used in a WHERE clause to search for a specified pattern in a column.

22. What is the difference between a view and a table?

Answer: A table is a physical storage structure that stores data, while a view is a virtual table derived from one or more tables, presenting a way to represent the result of a stored query.

23. What is a self-join?

Answer: A self-join is a regular join, but the table is joined with itself. It is often used when a table has a foreign key that references its own primary key.

24. Explain the purpose of the SQL IN operator.

Answer: The IN operator is used in a WHERE clause to filter the result set based on a specified list of values.

25. What is a subquery?

Answer: A subquery is a query nested inside another query. It can be used to retrieve data that will be used in the main query as a condition to further restrict the data to be retrieved.

26. What is the purpose of the SQL GROUP BY and COUNT() functions?

Answer: The GROUP BY clause is used to group rows based on specified columns, and COUNT() is an aggregate function that counts the number of rows in each group.

27. Explain the difference between a DELETE and TRUNCATE statement.

Answer: DELETE is used to remove rows from a table based on a condition, while TRUNCATE removes all rows from a table and is faster but cannot be rolled back.

28. What is a cross join?

Answer: A cross join, or Cartesian join, returns the Cartesian product of the sets of rows from the joined tables. It results in every combination of rows from both tables.

29. What is the purpose of the SQL BETWEEN operator?

Answer: The BETWEEN operator is used in a WHERE clause to filter the result set based on a range of values.

30. Explain the purpose of the SQL UPDATE statement.

Answer: The UPDATE statement is used to modify the existing records in a table. It allows you to update specific columns with new values.

31. What is DDL and DML? What is the difference?

Answer: DDL (Data Definition Language) is used to define the structure of the database (e.g., CREATE TABLE, ALTER TABLE, DROP TABLE).

DML (Data Manipulation Language) is used to manipulate the data within the database (e.g., INSERT, UPDATE, DELETE, SELECT).

32. What is the difference between Truncate vs Delete vs Drop?

Answer: **Truncate** removes all rows from a table quickly. It cannot be rolled back.

Delete removes specific rows from a table based on a condition. It can be rolled back.

Drop removes the entire table and its structure from the database. It cannot be rolled back.

33. When is Group By used?

Answer: The GROUP BY clause is used with aggregate functions (like SUM, AVG, COUNT) to group rows based on one or more columns and perform calculations within each group.

34. What is the difference between Where clause vs. Having clause?

Answer: Where clause filters rows before grouping and aggregation.

Having clause filters groups after grouping and aggregation.

35. What are Scalar and Aggregate functions in SQL?

Answer: Scalar functions return a single value for each row (e.g., UPPER, LOWER, SUBSTRING).

Aggregate functions return a single value for a group of rows (e.g., SUM, AVG, COUNT, MAX, MIN).

36. What does coalesce function do in SQL?

Answer: The COALESCE function returns the first non-null value in a list of arguments.

37. What are Constraints? How many Constraints are there?

Answer: Constraints are rules that enforce data integrity in a database.

Some common constraints include PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK.

38. How many primary key and unique key can be in a table?

Answer: A table can have only one PRIMARY KEY constraint.

A table can have multiple UNIQUE constraints.

39. What are Joins in SQL. Types of Joins in SQL?

Answer: Joins combine data from two or more tables based on a related column.

Common types include INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN, and CROSS JOIN

40. Difference between UNION and UNION ALL.

Answer: UNION combines the results of two or more SELECT statements, removing duplicate rows.

UNION ALL combines the results of two or more SELECT statements, including duplicate rows.

41. What is the order of execution in SQL Code?

Answer: The general order of execution is: FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY.

42. What are Window Functions or Analytical functions?

Answer: Window functions perform calculations across a set of rows related to the current row, such as ranking, partitioning, and calculating running totals

43. What is the diff between Rank & Dense Rank Function?

Answer: Rank assigns a rank to each row within a partition, with gaps for ties.

Dense Rank assigns a rank to each row within a partition, without gaps for ties.

44. What is View? How it is different from Table?

Answer: A view is a virtual table that is created from the result of an SQL query. It doesn't physically store data but provides a dynamic view of the underlying data.

45. What is Index?

Answer: An index is a data structure that improves the speed of data retrieval by creating a sorted order for specific columns in a table.

46. What is CTE? How it is useful?

Answer: A Common Table Expression (CTE) is a temporary named result set that you can define within a SELECT, INSERT, UPDATE, or DELETE statement.

CTEs can make complex queries more readable and maintainable.

47. What is difference between SQL vs. PLSQL?

Answer: SQL (Structured Query Language) is used to interact with databases, primarily for data retrieval and manipulation.

PL/SQL (Procedural Language/SQL) is an extension of SQL that adds procedural programming capabilities to SQL, such as loops, conditional statements, and variables.

48. What is the use case of Cross Join?

Answer: CROSS JOIN is used to create a Cartesian product (all possible combinations) of rows from two tables. It is often used in conjunction with other clauses (like WHERE) to filter the results

49. Explain the ACID properties of transactions and their importance.

Answer: ACID stands for Atomicity, Consistency, Isolation, and Durability. These properties ensure the reliability and integrity of database transactions.

Atomicity: All-or-nothing execution of a transaction.

Consistency: Maintains database integrity after a transaction.

Isolation: Transactions are independent and do not interfere with each other.

Durability: Committed changes are permanent and survive system failures.

50. Describe the difference between a Primary & Foreign Key.

Answer: Primary Key: A unique identifier for each row in a table. It cannot contain null values and must be unique for every record.

Foreign Key: A column in one table that references the Primary Key of another table. It establishes a relationship between the two tables.

51. How do Indexes work in a database?

Answer: Indexing stores a sorted list of values for one or more columns, along with pointers to the actual data rows.

Benefits: Significantly improves query performance, especially for searches, joins, and sorting. Enables faster data retrieval and updates.

Drawbacks: Increased storage space required for the index. Can slow down data insertion and updates as the index needs to be maintained.

52. Explain the concept of normalization in database design.

Answer: Normalization is the process of organizing data in a database to minimize redundancy and improve data integrity. It involves breaking down large tables into smaller, more manageable tables with well-defined relationships.

Why is it important: Improves data consistency and reduces the risk of data anomalies. Makes data easier to maintain and update. Improves query performance.

53. Explain Subqueries & its different types with examples.

Answer: Subqueries are nested SELECT statements within another SQL statement.

Scalar Subquery: Returns a single value (e.g., used in WHERE clause to compare a column value with the result of the subquery).

Correlated Subquery: References a column from the outer query within the subquery.

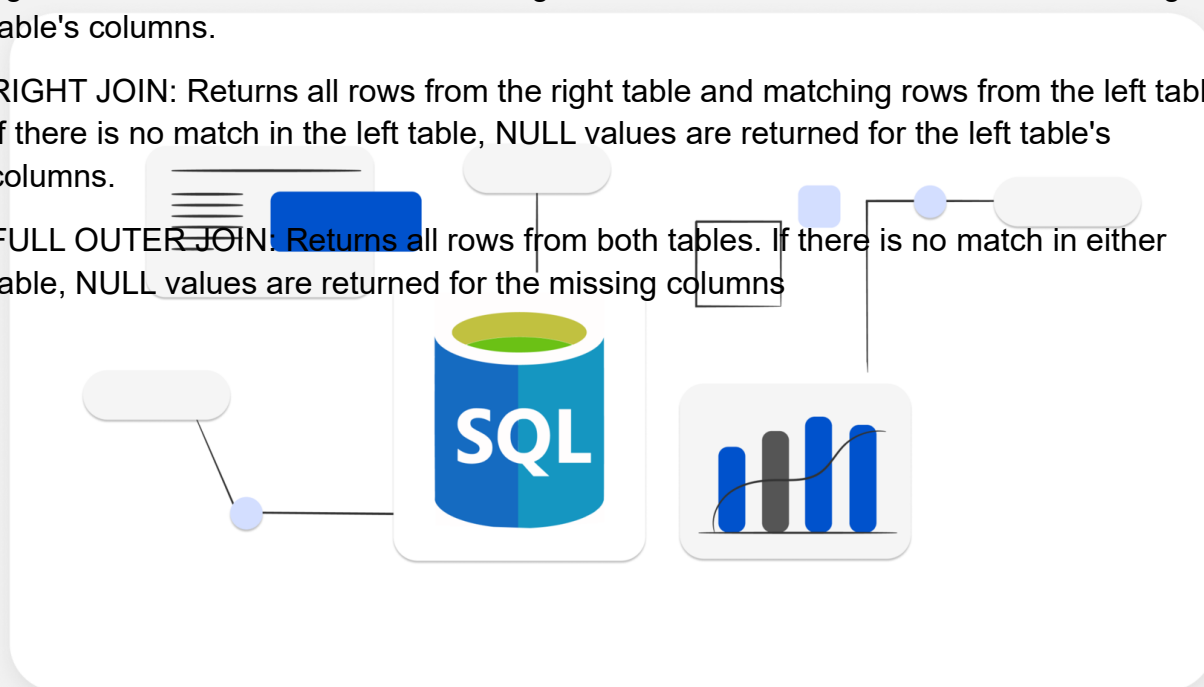
Multiple-Row Subquery: Returns multiple rows (e.g., used with IN or EXISTS operators)

54. Explain left join, right join, and full outer join with examples.

Answer: LEFT JOIN: Returns all rows from the left table and matching rows from the right table. If there is no match in the right table, NULL values are returned for the right table's columns.

RIGHT JOIN: Returns all rows from the right table and matching rows from the left table. If there is no match in the left table, NULL values are returned for the left table's columns.

FULL OUTER JOIN: Returns all rows from both tables. If there is no match in either table, NULL values are returned for the missing columns



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